PROFESSIONAL SUMMARY:

- Experienced professional with **9+** years of expertise as a **Data Scientist, AI, and ML Engineer**, skilled in developing and implementing cutting-edge **machine learning models**, **AI-**powered solutions, and data-centric strategies to address intricate business problems and foster innovation.
- Proficient in **Python** and **R** for **statistical analysis, machine learning model development**, and data visualization.
- Experienced in deep learning frameworks like **TensorFlow, Keras, and PyTorch** for building advanced neural network architectures.
- Skilled in utilizing **Scikit-learn** and **Apache Spark's MLLib** for implementing regression, classification, clustering, and dimensionality reduction techniques.
- Expertise in deploying and managing scalable **machine learning models** using **AWS EC2 instances,** optimizing resources for high-performance computing.
- Strong background in leveraging **AWS RDS** for setting up, managing, and scaling relational databases to support analytics applications.
- Skilled in implementing robust security practices using **AWS IAM**, ensuring controlled access and compliance for machine learning models and data pipelines.
- Proven expertise in designing **end-to-end machine learning pipelines**, automating model training, validation, and deployment for scalable AI solutions.
- Strong experience in **model optimization and hyperparameter tuning**, improving accuracy and performance in production environments.
- Skilled in **real-time data streaming** and **event-driven architectures**, utilizing **Apache Kafka** and **AWS Kinesis** for high-frequency data processing.
- Experienced in **MLOps practices**, integrating CI/CD pipelines for machine learning models using **Azure DevOps, GitHub Actions, and Jenkins**.
- Strong command of **time-series forecasting techniques**, applying advanced algorithms for predictive analytics in financial and operational domains.
- Hands-on experience in **explainable AI (XAI) and model interpretability**, utilizing **SHAP**, **LIME**, and **Fairness Indicators** for transparent AI solutions.
- Experience in **developing reinforcement learning algorithms** for decision automation and optimization problems.
- Strong foundation in **causal inference techniques**, enabling data-driven decision-making beyond correlation-based models.
- Experience in **managing data versioning and lineage tracking** using **MLflow and DVC**, ensuring reproducibility of AI experiments.
- Strong command of **algorithmic trading and financial modeling**, leveraging AI for risk assessment and market predictions.
- Expertise in database management using **PostgreSQL, MySQL, SQL Server, and MongoDB**, ensuring optimal data storage and retrieval and Adept at cloud computing platforms including **AWS** and **Azure** for deploying scalable machine learning and **AI** solutions.
- Hands-on experience with **big data tools** such as **Apache Spark, PySpark, Hadoop, HIVE, MapReduce**, and **HDFS** for processing large-scale datasets.
- Proficient in b tools like **AWS Glue, Azure Data Factory, SSIS, Informatica, and Talend** for seamless data integration workflows.
- Strong knowledge of **data manipulation** and using **Pandas, NumPy, SAS, and RStudio** to derive actionable insights.
- Expertise in creating interactive dashboards and reports using **Tableau and Power BI** to visualize key performance metrics.
- Skilled in **NLP tools** such as **NLTK, SpaCy, Gensim, and Azure Cognitive Services** for building natural language understanding solutions.
- Skilled in developing **AI-driven chatbots** and **virtual assistants**, leveraging NLP frameworks like **Dialogflow**, **Rasa**, and **Microsoft Bot Framework.**
- Experienced in computer **vision techniques** using **OpenCV** for image recognition and processing applications.
- Proficient in utilizing **AWS S3** for secure and efficient storage, handling large datasets and backups for data-driven solutions.
- Experienced in building architectures with **AWS Lambda** to automate workflows and integrate with various data processing services.
- Knowledgeable in containerization and orchestration using Docker and Kubernetes to streamline model deployment processes and Strong version control expertise with **Git** and **SVN** to manage collaborative development projects effectively.
- Familiar with **Azure ML Pipelines**, **AKS, AWS SageMaker**, and **Azure Databricks** for automated **ML** workflows and scalable solutions.
- Specialized in developing **recommendation systems, predictive analytics models**, and **ERP/CRM AI** solutions to address complex business challenges.

TECHNICAL SKILLS:

Programming Languages	Python, R, SQL, JavaScript, Bash
Machine Learning Frameworks	TensorFlow, Keras, PyTorch, Scikit-learn, MLlib, Apache Spark
Deep Learning	CNNs, RNNs, Neural Networks, Image Recognition, Text Classification, Sentiment Analysis
Big Data & Distributed Systems	Apache Spark, PySpark, Hadoop, HDFS, HIVE, MapReduce, Kafka
Cloud Platforms	AWS, Azure
Data Storage & Management	PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Netezza
Data Integration & ETL Tools	AWS Glue, Azure Data Factory, SSIS, Informatica, Talend, Apache Kafka, Apache Spark
Version Control	Git, SVN, Azure DevOps, GitHub, Jenkins, Docker
Data Visualization	Tableau, Power BI, Matplotlib, Seaborn, R Studio
Containerization & Orchestration	Docker, Kubernetes, AWS Elastic Beanstalk, AKS
NLP & Text Processing	NLTK, SpaCy, Gensim, Azure Cognitive Services
Computer Vision	OpenCV, Image Recognition, Object Detection, Anomaly
Statistical Analysis & Testing	Pandas, NumPy, SAS, Hypothesis Testing, A/B Testing
Model Deployment & Automation	Azure DevOps, Docker, Kubernetes, Azure Resource Manager, AWS Lambda, AWS S3, Elastic Beanstalk
AI Model Development	TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, NLP, Deep Learning, Model Optimization

PROFESSIONAL EXPERIENCE:

Client: American Express, New York, NY. **Apr 2023 - Present**

Role: Data Scientist/AI/ ML Engineer
Responsibilities:

- Designed, trained, and deployed machine learning models using **Scikit-learn, TensorFlow, and PyTorch** for **predictive analytics and classification tasks.**
- Built scalable **data pipelines** leveraging **Apache Spark, EMR, and Hadoop** to process terabytes of structured and unstructured data efficiently.
- Analyzed large datasets from **HDFS** and **Hive** to uncover actionable insights and support data-driven decision-making.
- Utilized **Azure DevOp**s for version control, **continuous integration, and deployment (CI/CD)** of machine learning models and data pipelines.
- Leveraged **TensorFlow** to design, train, and deploy deep learning models, improving the efficiency of **AI** solutions in production.
- Utilized **Azure Machine Learning** services for **model training, deployment, and monitoring** in a cloud-based environment and engineered advanced data transformations with **Pandas** and **NumPy** to preprocess data for machine learning workflows.
- Developed and deployed **machine learning models** using **Python, Scikit-learn,** and **TensorFlow** for predictive analytics and Utilized **Hadoop and Spark** for distributed data processing, ensuring scalable and efficient **data handling.**
- Built interactive dashboards and data visualizations using **Power BI** for business insights.
- Implemented **Natural Language Processing (NLP) techniques** for **text classification, sentiment analysis,** and **language modeling.**
- Developed and deployed scalable **AI models in Azure Kubernetes Service (AKS**) for high-availability containerized applications.
- Created compelling **data visualizations** using **Matplotlib** and **Seaborn** to communicate findings to **technical** and **non-technical stakeholders**.
- Leveraged **Matplotlib** and **Seaborn** to **visualize data distributions, trends, and model performance**, presenting findings to stakeholders.
- Developed and maintained **AI** applications using Keras for building neural network architectures, optimizing models for real-time processing.

- Employed **Azure Resource Manager (ARM) templates** to automate infrastructure deployment, ensuring consistency and scalability for data science projects.
- Applied **Azure Logic Apps** to streamline and automate workflows between **data sources, analytics services, and reporting tools** and Managed **Azure Key Vault** to secure and store sensitive data, such as **API keys and model credentials**, ensuring compliance with best practices.
- Monitored data processing and model performance using **Azure Monitor,** providing actionable insights and optimizing resources for efficient operation.
- Designed and implemented **machine learning pipelines**, **automating model training**, **validation**, and **deployment** in production environments.
- Developed **AI models** using **machine learning techniques** to solve complex problems and optimize processes and applied **deep learning algorithm**s to enhance **AI capabilities** for image recognition and data classification.
- Worked with **SQL and Hadoop** to manage and analyze large-scale datasets, optimizing queries and storage strategies for better performance.
- Integrated **Apache Spar**k with **machine learning models** to process **big data** efficiently and train models at scale and collaborated with cross-functional teams to integrate **machine learning models** into business applications and deliver actionable insights.
- Conducted **sentiment analysis** and **topic modeling** using **Natural Language Processing (NLP) techniques** and libraries.
- Developed, containerized, and deployed **AI/ML solutions** using **Docker** and orchestrated workflows with Kubernetes.
- Utilized **Azure cloud services** for **developing, deploying, and monitoring machine learning models** in production.
- Designed and maintained **logical** and **physical data models** with **Erwin Data Modeler** to support analytics and reporting and built and managed **ETL pipelines** using **Informatica** to integrate data from multiple sources into a centralized repository.
- Implemented computer **vision models** with **OpenCV** for image recognition, object detection, and anomaly detection and Conducted **A/B testing** and **statistical analysis** to optimize product features and validate **machine learning outcomes.**
- Leveraged **Power BI** to create **dynamic dashboards, providing stakeholders** with real-time insights into operational metrics.
- Collaborated with cross-functional teams using **Jira** and **Confluence** to manage projects and document workflows efficiently and designed **big data solutions** integrating **MongoDB** with **Apache Spark**.

Environment: Scikit-learn, TensorFlow, PyTorch, Apache Spark, EMR, Hadoop, HDFS, Hive, Azure Pandas, NumPy, Power BI, Natural Language Processing (NLP), Matplotlib, Seaborn, Keras, Docker, Kubernetes, SQL, Informatica, Erwin Data Modeler, MongoDB, OpenCV, A/B Testing, Jira, Confluence.

Client: Allstate, Northbrook, IL. **Jan 2020 – Mar 2023**

Role: Data Scientist/ AI/ML Engineer

Responsibilities:

- Designed and implemented **machine learning models** using **TensorFlow, PyTorch, and scikit-learn** to solve complex business problems and drive data-driven decision-making.
- Orchestrated and monitored distributed workloads using **Apache Spark and HIVE**, ensuring scalability and fault tolerance.
- Developed scalable data pipelines and preprocessing workflows utilizing **Apache Spark, MLlib,** and **Pandas** for feature engineering and data transformations.
- Built and deployed **ML-based solutions leveraging OpenCV** for computer vision tasks such as object detection and image classification and Automated **data ingestion** and transformation processes using **Apache Kafka** to enable real-time analytics for dynamic systems.
- Engineered robust big data solutions on **Hadoop** and **Hive**, optimizing query performance for large-scale datasets.
- Managed and queried structured and unstructured data in **SQL Server, Netezza, and MongoDB**, ensuring seamless integration with **ML** workflows.
- Conducted **data cleaning** and wrangling with **Pandas** and **NumPy** to prepare raw data for analysis.
- Applied **deep learning algorithms** using **Keras and PyTorch** to create advanced models for image recognition and classification.
- Utilized **natural language processing (NLP)** to improve sentiment analysis and chatbots.
- Leveraged computer vision technologies in **AI** projects to enable automated visual inspection and object detection.
- Developed and **tested data pipelines** with **Apache Kafka** for real-time data streaming and integration.
- Performed statistical analysis and hypothesis testing to guide data-driven decision-making.
- Implemented **machine learning algorithms** to automate forecasting, recommendation systems, and anomaly detection.

- Developed machine learning models using **Python, R, TensorFlow**, and **PyTorch** to solve complex business problems and improve decision-making.
- Utilized **AWS EC2** for scalable data processing and model training, ensuring high-performance computation for large datasets and Leveraged **AWS S3** for secure and efficient storage of raw data, processed files, and model outputs.
- Implemented **AWS Lambda** for serverless computing, automating data preprocessing tasks and real-time data transformations in **data science** workflows.
- Managed relational and non-relational databases using **Amazon RDS** and **DynamoDB**, ensuring optimized storage solutions for structured and unstructured data.
- Utilized **Scikit-learn** for **implementing machine learning algorithms** and **techniques** such as **regression**, **classification, and clustering.**
- Preprocessed and cleaned large datasets using **Pandas and NumPy,** ensuring **data quality** and compatibility for **model training.**
- Built and optimized **deep learning architectures**, including **CNNs and RNNs**, using **Keras** and **TensorFlow** for image and text processing tasks.
- Applied **Natural Language Processing (NLP) techniques** to process and analyze textual data, improving data-driven insights.
- Created and maintained **SQL queries** for data extraction, transformation, and analysis from large datasets.
- Designed interactive dashboards and visualizations in **Power BI** to communicate **machine learning** outcomes and trends effectively to stakeholders.
- Configured and maintained **AWS IAM** roles and policies, ensuring secure access to data and services for data science applications. Monitored model performance and logs with **AWS CloudWatch**.
- Integrated messaging services such as **AWS SNS and AWS SQS** to create reliable communication between data pipelines and model-serving systems.
- Containerized **ML applications** using **Docker** to ensure environment consistency across development and **production stages.**
- Utilized **R Studio** for **statistical modeling, hypothesis testing, and reporting** to validate **ML model performance** and implemented continuous integration and version control workflows using **Git** to maintain a streamlined codebase for **ML projects**.

Environment: TensorFlow, PyTorch, scikit-learn, Apache Spark, HIVE, Apache Kafka, OpenCV, Pandas, MLlib, Keras, NumPy, AWS, Docker, R Studio, Git, Power BI, SQL Server, Netezza, MongoDB, Hadoop, Keras, CNN, RNN, NLP.

Client: Fortis Healthcare, Gurgaon, India. **Oct 2017 – Nov 2019**

Role: Data Scientist

Responsibilities:

- Developed predictive models using **SVM, Random Forest, and Boosted Trees** to optimize business outcomes and improve decision-making processes.
- Implemented **Regression techniques** to identify **key variables** and predict future trends based on historical data.
- Built and fine-tuned **Neural Networks** using **TensorFlow** to solve complex problems, such as image recognition and text classification and Engineered advanced Data Imputation methods to handle missing data, ensuring the integrity and reliability of datasets.
- Utilized **Azure Machine Learning Studio** to design, deploy, and **manage machine-learning models** for predictive analytics and real-time data processing.
- Integrated **Azure SQL Database** for storing and querying large datasets, optimizing performance and ensuring data integrity.
- Leveraged **Azure Blob Storage** to efficiently manage and process big data, Implemented **Azure Functions** to automate data ingestion pipelines and enhance real-time data processing capabilities for machine learning applications.
- Conducted A**/B testing** and **analysis** for product optimization using **statistical** and **machine learning** techniques.
- Created interactive dashboards in **Tableau** to **visualize trends, KPIs**, and **analytical insights** for stakeholders and Utilized **Excel** for quick **data manipulation**, **exploratory analysis, and model validation.**
- Employed **Seaborn** and **Matplotlib** to create clear, insightful **visualizations** that support data-driven decision-making.
- Designed and implemented **Feature Engineering strategies** to enhance model performance and accuracy.
- Collaborated with cross-functional teams to deploy real-time **machine learning solutions** utilizing **Azure ML, Docker, and Kafka** for end-to-end automation.
- Optimized model training processes with **Stochastic Gradient Descent**, reducing computational overhead and improving efficiency and wrote efficient **SQL queries** for data extraction, transformation, and analysis from large relational databases.

- Processed and analyzed **big data** using **Hadoop**, enabling scalable data storage and distributed processing.
- Conducted end-to-end **machine learning pipelines**, including **data preprocessing, model selection,** and **hyperparameter tuning.**
- Collaborated with cross-functional teams to integrate data science models into production **systems using Git** and **Docker** and Provided actionable insights by designing and delivering comprehensive reports and visualizations to stakeholders.

Environment: SVM, Random Forest, Boosted Trees, Regression Techniques, Clustering Analysis, Neural Networks, Data Imputation, Azure, A/B Testing, Tableau, Excel, Seaborn, Matplotlib, Feature Engineering, Docker, Kafka, SGD, SQL, Hadoop, Git, Machine Learning Pipelines, Hyperparameter Tuning.

Client: Aditya Birla Retail, Mumbai, India. **Jul 2015 – Sep 2017**

Role: Data Scientist
Responsibilities:

- Designed and implemented **data preprocessing** workflows using **Python, R, and SQL** to clean, transform, and standardize large datasets for machine learning models.
- Built and optimized data pipelines with **Spark and Hive** to process massive datasets efficiently, ensuring high data availability and reliability.
- Developed and deployed **machine learning models** in production environments using **PyTorch**, focusing on improving predictive accuracy and scalability.
- Leveraged **AWS services** such as **S3, EC2, and SageMaker** for data storage, computation, and deployment of **machine learning models.**
- Conducted feature engineering to extract meaningful variables from **raw data,** enhancing model performance and interpretability.
- Analyzed complex datasets and visualized insights using **Tableau**, creating interactive dashboards to support business decision-making.
- Collaborated with cross-functional teams to design end-to-end data solutions, integrating machine-learning models with existing systems.
- Created and maintained **SQL queries** for data extraction, analysis, and integration with machine learning workflows.
- Applied **advanced statistical methods** and **machine learning techniques** in **R** to uncover actionable insights from structured and unstructured data.
- Utilized **Microsoft Office Suite** to document project workflows, prepare presentations, and communicate findings to non-technical stakeholders effectively.

Environment: Python, R, SQL, Spark, Hive, PyTorch, AWS, Tableau, Feature Engineering, Machine Learning, Data Preprocessing, Data Visualization, SQL Queries, Statistical Methods, Microsoft Office Suite.

<u>EDUCATION:</u> BTech in Computer Science and Engineering, June 2011 - May 2015 at JNTUH, India.